4MA1_2H_ch3_sequences_functions_graphs

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1. 4MA1/2H_Winter_2024_Q19

A curve has equation $y = \mathrm{f}(x)$

There is only one minimum point on the curve.
The coordinates of this minimum point are (5, 4)

Write down the coordinates of the minimum point on the curve with equation

(i) $y = \mathrm{f}(x+5)$

(.................... , ...)
(1)

(ii) $y = 3\mathrm{f}(x)$

(.................... , ...)
(1)

(iii) $y = \mathrm{f}(x) - 7$

(.................... , ...)
(1)

2. 4MA1/2H_Winter_2024_Q18

The straight line **L** is perpendicular to the straight line with equation $2x + y = 9$ and passes through the point with coordinates (8, 11)

Find an equation for **L**
Give your answer in the form $y = mx + c$

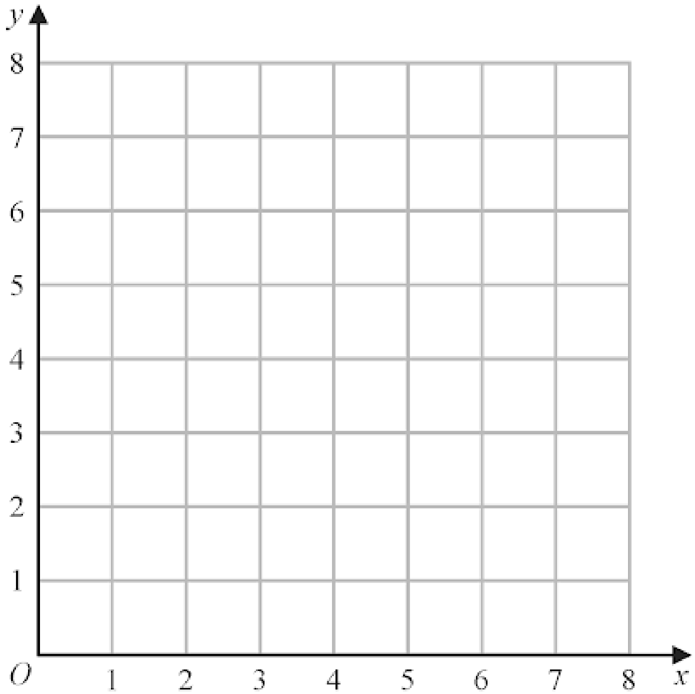
4 marks

3. 4MA1/2H_Winter_2024_Q7

(a) On the grid, draw the straight line with equation

(i) $x = 3$ (ii) $y = 1$ (iii) $x + y = 7$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$$x \geqslant 3 \qquad y \geqslant 1 \qquad x + y \leqslant 7$$

Label the region **R**

(1)

4. 4MA1/2HR_Summer_2024_Q23

A curve has equation $y = \mathrm{f}(x)$
The coordinates of the minimum point on this curve are $(6, -3)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = \mathrm{f}(x) + 10$

(.................... , ...)
(1)

(ii) $y = \mathrm{f}(3x)$

(.................... , ...)
(1)

5. 4MA1/2HR_Summer_2024_Q17

$\mathrm{f}(x) = \dfrac{x}{2x - 4}$ $\quad$ $\mathrm{g}(x) = 3x + 1$

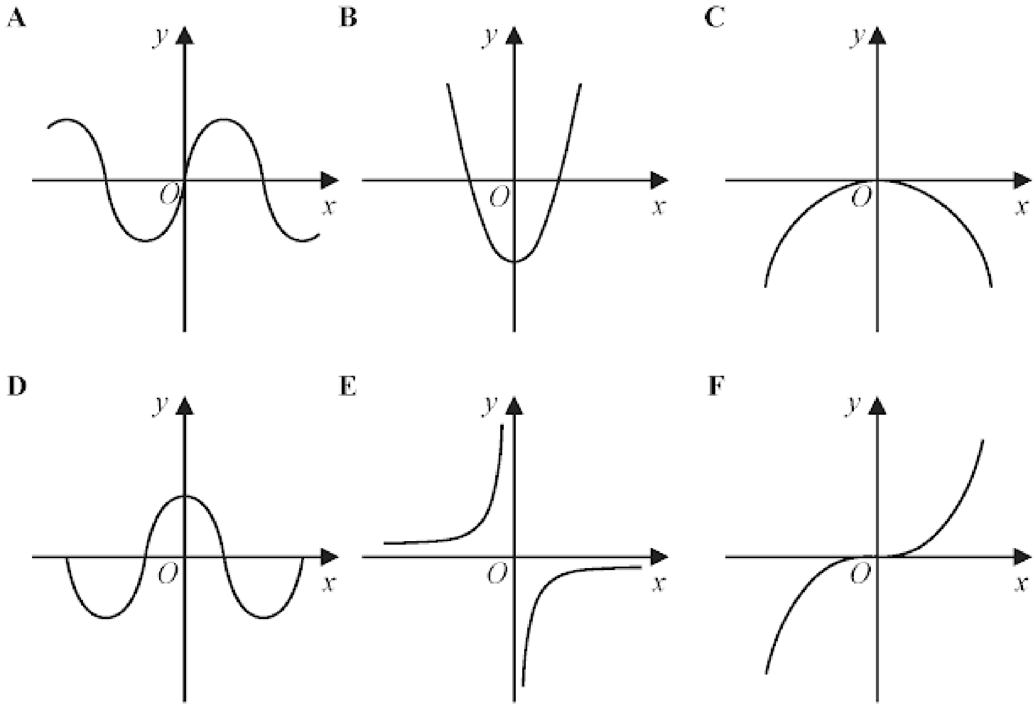
Given that $\mathrm{fg}(k) = 2$

work out the value of k

(3)

6. 4MA1/2HR_Summer_2024_Q16

Here are six graphs.



Write down the letter of the graph that could have the equation

(i) $y = -\dfrac{1}{x}$

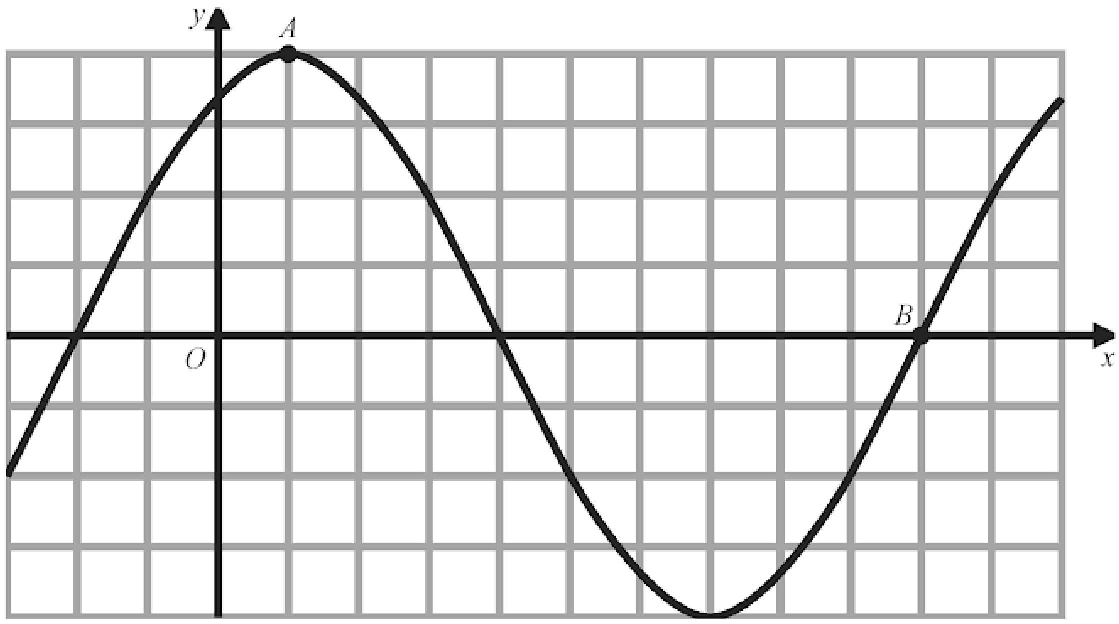
(1)

(ii) $y = \sin x°$

(1)

7. 4MA1/2H_Summer_2024_Q25

The diagram shows a sketch of the graph of $y = 2\sin(x + 60)°$



(i) Find the coordinates of the point A

(.................... , ...)

(1)

(ii) Find the coordinates of the point B

(.................... , ...)

(1)

8. 4MA1/2H_Summer_2024_Q22

The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(.................... , ...) (.................... , ...)

(5)

9. 4MA1/2H_Summer_2024_Q18

The straight line **P** has equation $5y + 2x = 7$

Find the gradient of a straight line that is perpendicular to **P**

(2)

10. 4MA1/2H_Summer_2024_Q2

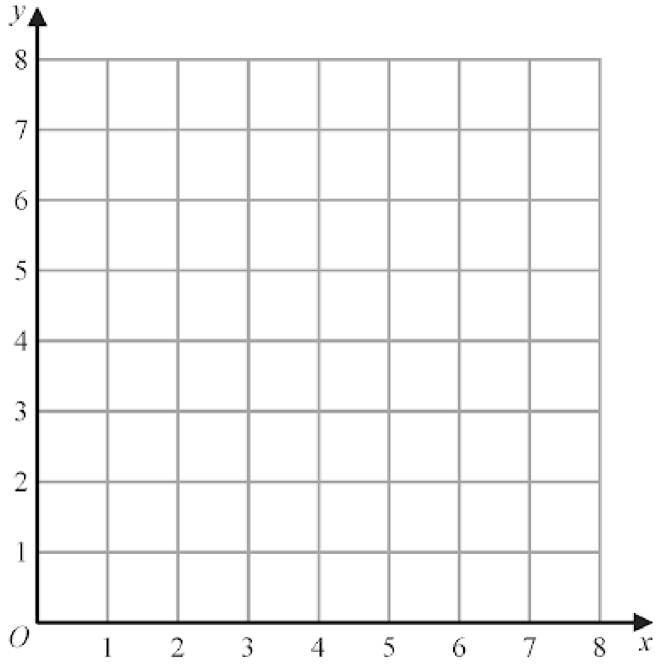
(a) On the grid, draw the straight line with equation

(i) $y = 2$

(ii) $x = 6$

(iii) $y = x + 1$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

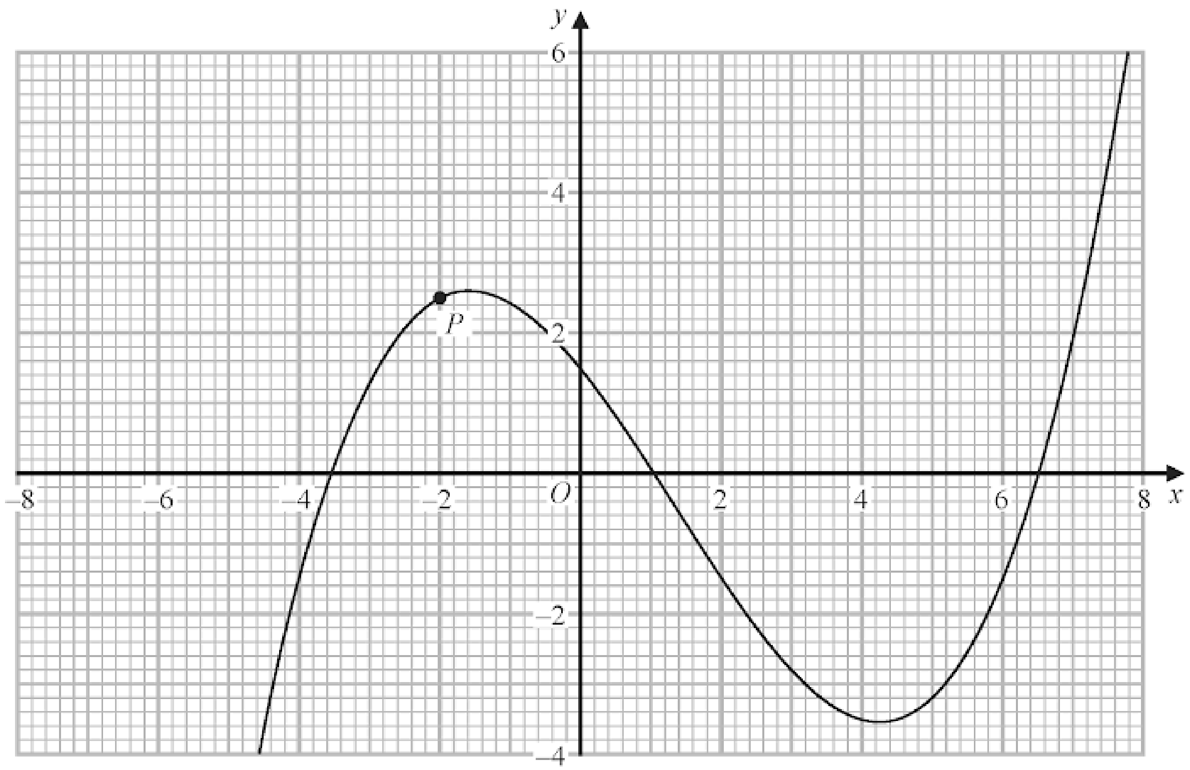
$$y \geqslant 2 \qquad x \leqslant 6 \qquad y \leqslant x + 1$$

Label the region **R**

(1)

11. 4MA1/2HR_Winter_2023_Q21

The diagram shows the graph of $y = f(x)$



The point P has x coordinate –2

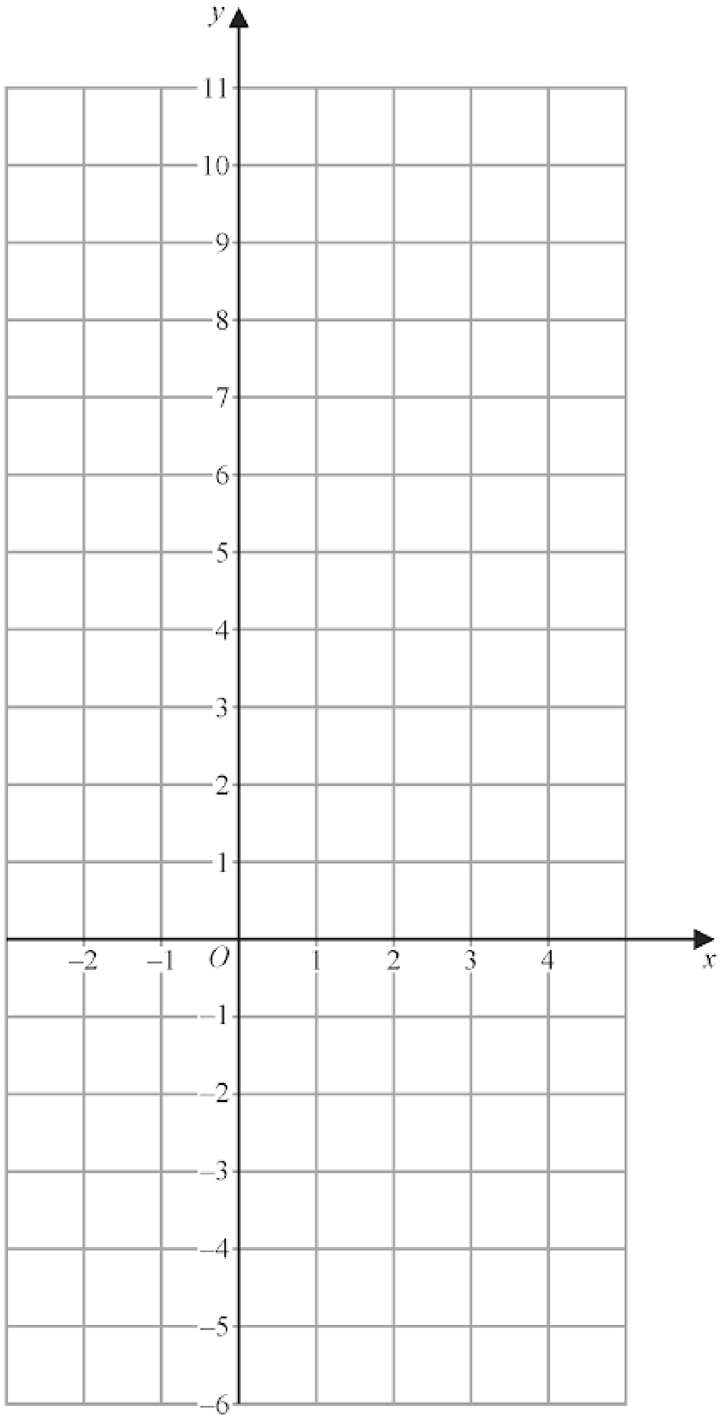
Use the graph to find an estimate for the gradient of the curve at P

....

3 marks

12. 4MA1/2HR_Winter_2023_Q5

On the grid, draw the graph of $5x + 2y = 10$ for values of x from -2 to 4



3 marks

13. 4MA1/2H_Winter_2023_Q25

The function f is such that $f(x) = 3x^2 - 12x + 7$ where $x \leqslant 2$

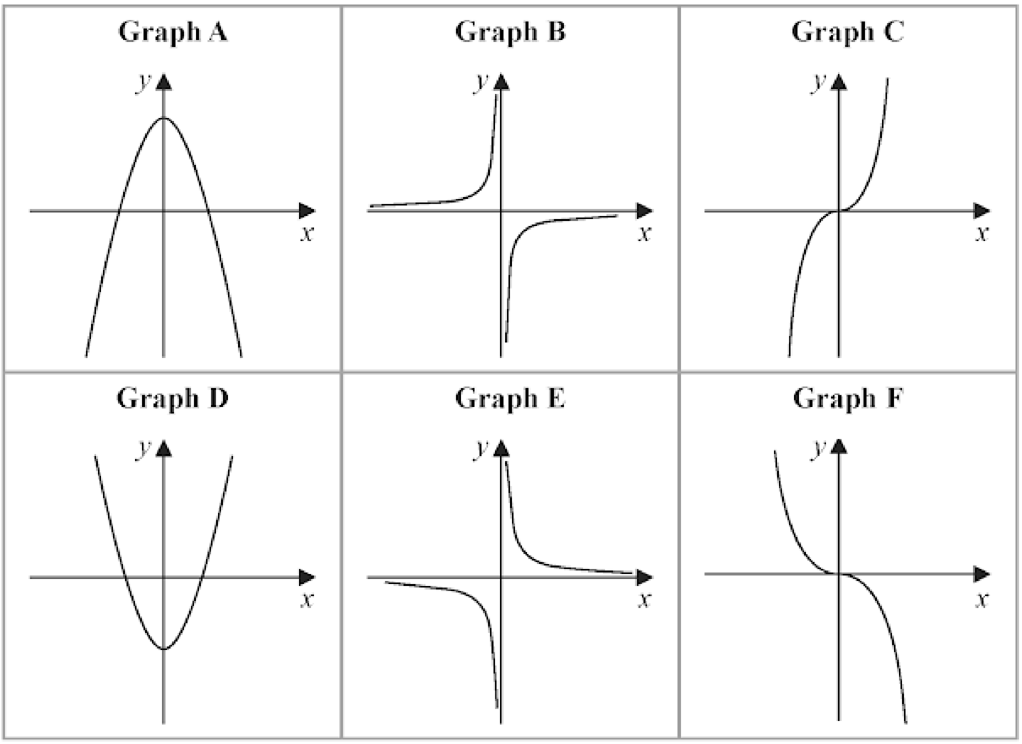
Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ ....

4 marks

14. 4MA1/2H_Winter_2023_Q11

Here are six graphs.



Complete the table below with the letter of the graph that could represent each given equation.
Write your answers on the dotted lines.

Equation	Graph
$y = -\dfrac{2}{x}$	....
$y = 5 - x^2$	....
$y = -2x^3$	....

3 marks

15. 4MA1/2H_Winter_2023_Q6

The points A and B are on a coordinate grid.

The coordinates of A are $(6, 4)$
The coordinates of B are $(17, j)$ where j is a constant.

The midpoint of AB has coordinates $(k, 15)$ where k is a constant.

Find the value of j and the value of k

$j =$ ....

$k =$ ....

3 marks

16. 4MA1/2HR_Summer_2023_Q25

The straight line with equation $y - 2x = 7$ is the perpendicular bisector of the line AB where A is the point with coordinates $(j, 7)$ and B is the point with coordinates $(6, k)$

Find the coordinates of the midpoint of the line AB
Show clear algebraic working.

17. 4MA1/2HR_Summer_2023_Q23

Here are the first three terms of an arithmetic sequence.

$$8p \qquad 7p - 3 \qquad 4p + 2$$

The sum of the first n terms of the sequence is -1914

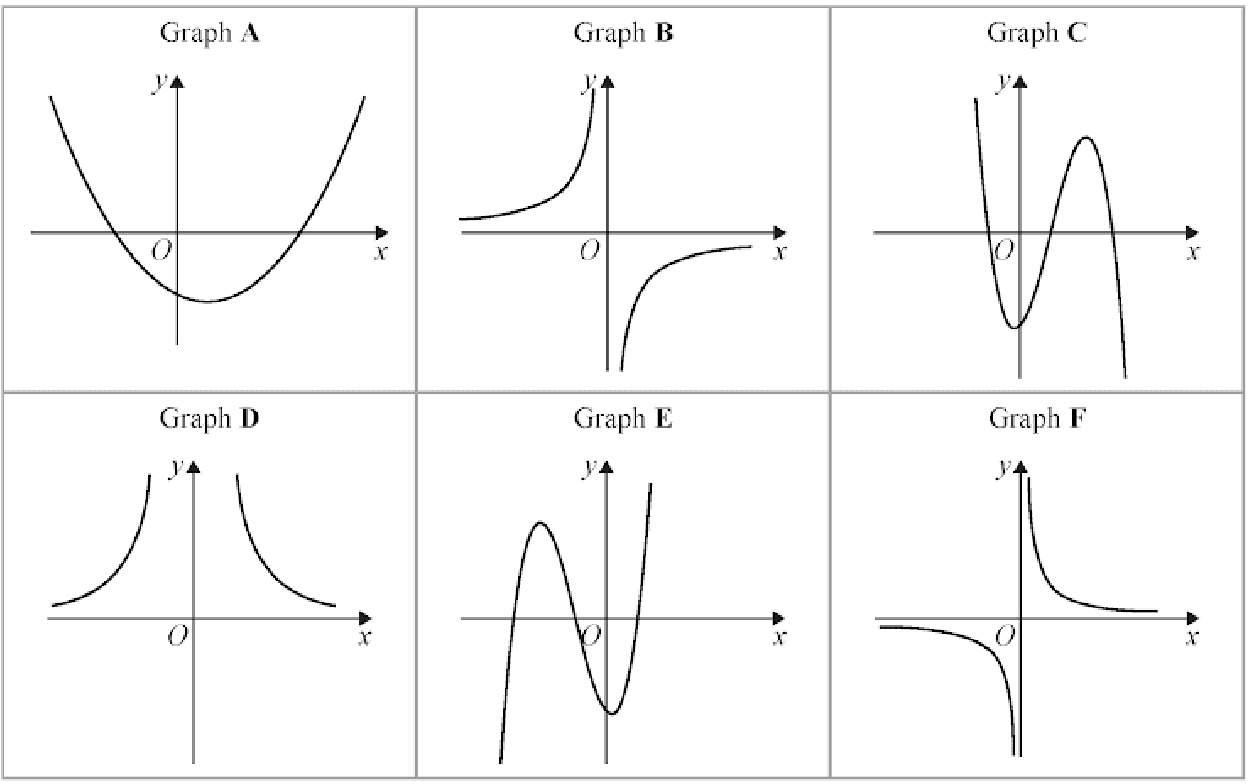
Work out the value of n
Show your working clearly.

$n =$ ....

5 marks

18. 4MA1/2HR_Summer_2023_Q12

Here are six graphs.



Write down the letter of the graph of

(a) $y = \dfrac{10}{x^2}$

....

(1)

(b) $y = x - 3 + 3x^2 - x^3$

....

(1)

(c) $y = -\dfrac{3}{x}$

....

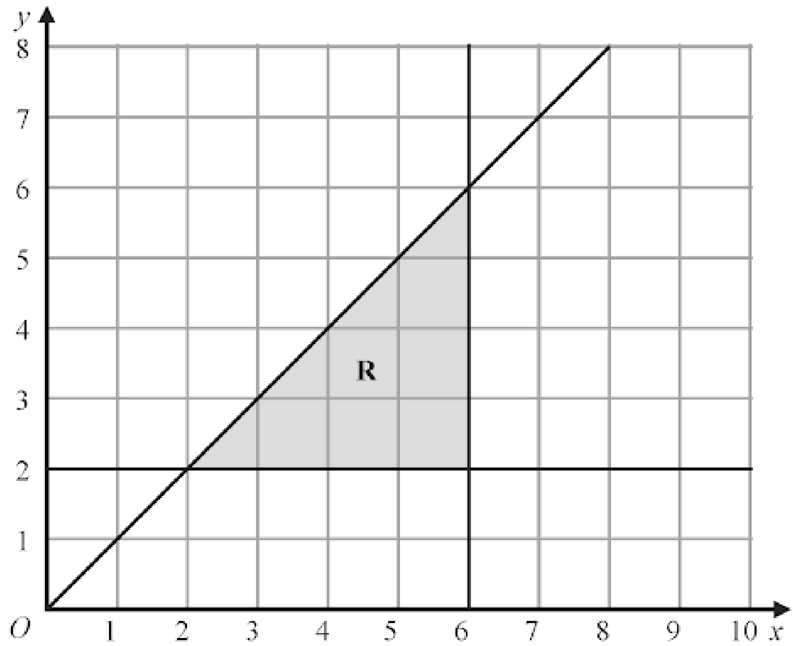
(1)

19. 4MA1/2HR_Summer_2023_Q8

(a) Solve $9 - 4x > 17$

....

(2)



(b) Write down the three inequalities that represent the shaded region **R**

....

....

....

(3)

20. 4MA1/2HR_Summer_2023_Q4

Here are the first four terms of an arithmetic sequence.

38 31 24 17

Find an expression, in terms of n, for the nth term of the sequence.

....

2 marks

21. 4MA1/2H_Summer_2023_Q24

$ABCD$ is a kite with $AB = AD$ and $CB = CD$

A is the point with coordinates $(-2, 10)$

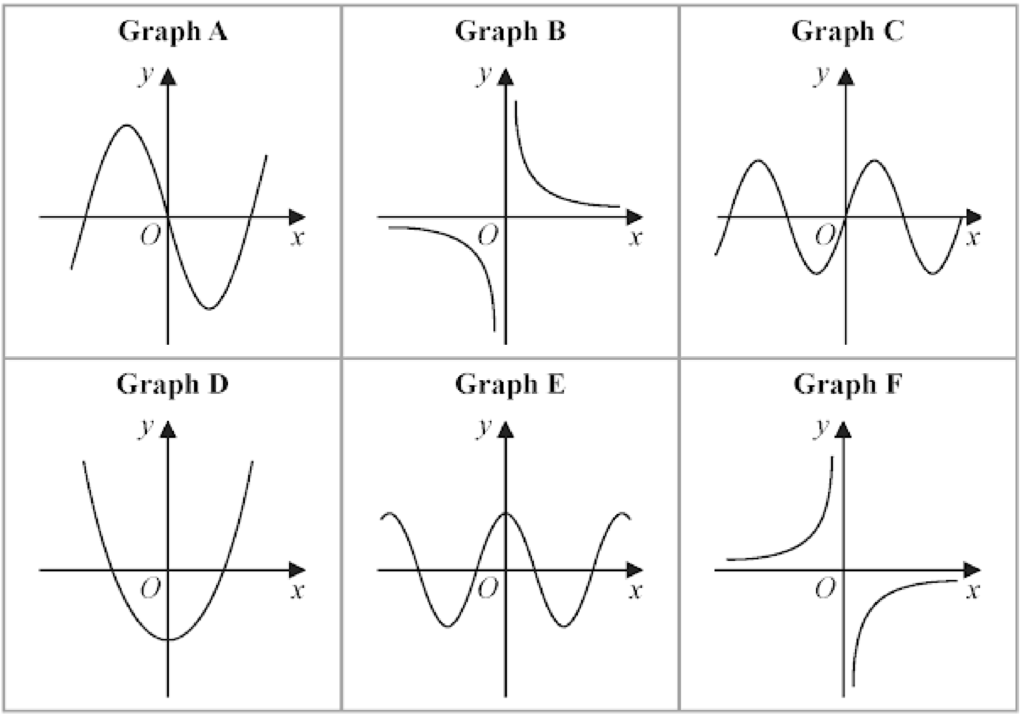
B is the point with coordinates $\left(-\frac{27}{5}, 4\right)$

C is the point with coordinates $(4, -5)$

Work out the coordinates of D

22. 4MA1/2H_Summer_2023_Q18

Here are 6 graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \sin x$	....
$y = -\dfrac{3}{x}$	....
$y = 4x^3 - 5x$	....

3 marks

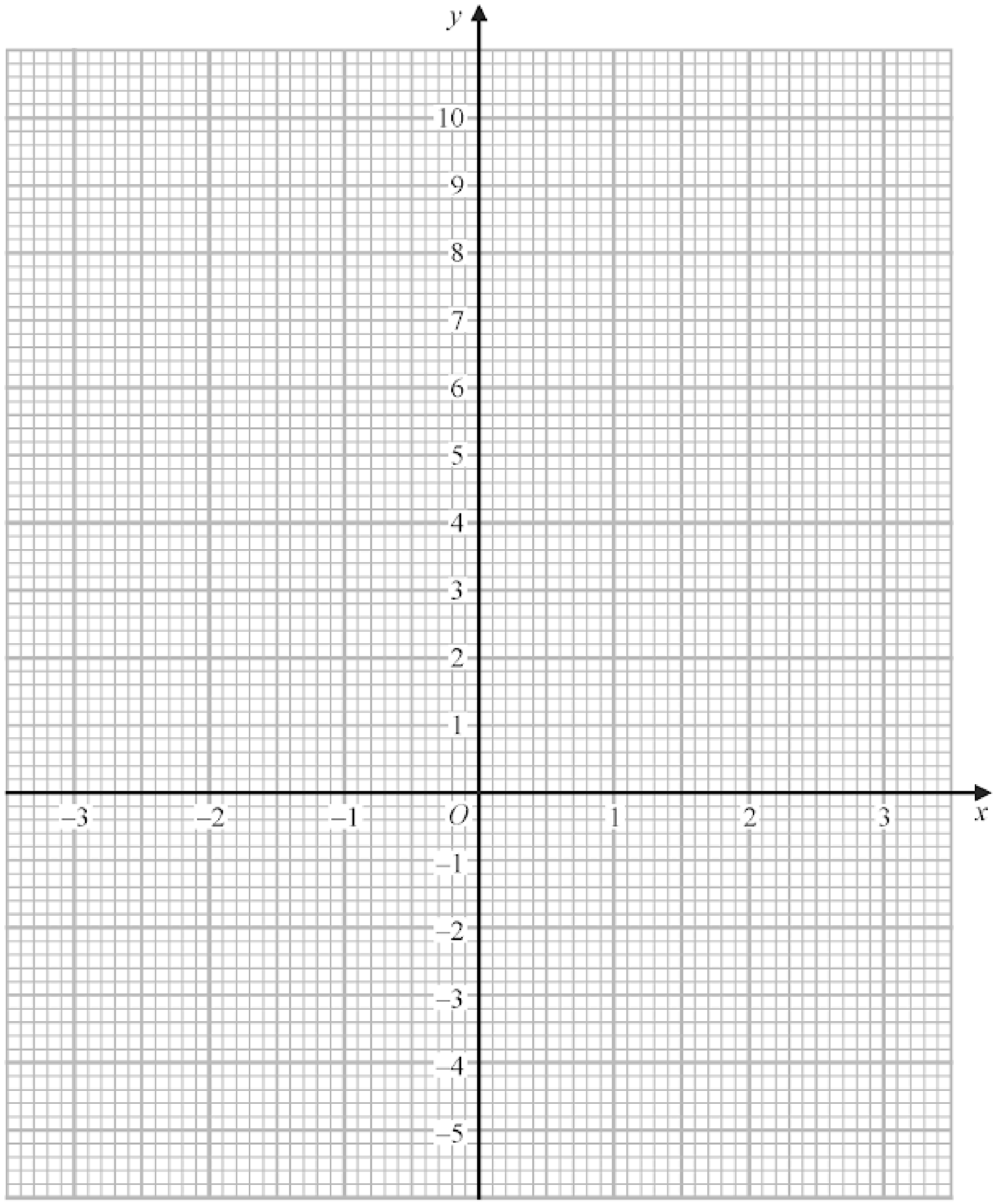
23. 4MA1/2H_Summer_2023_Q4

(a) Complete the table of values for $y = x^2 - x - 4$

x	-3	-2	-1	0	1	2	3
y		2			-4		

(2)

(b) On the grid below, draw the graph of $y = x^2 - x - 4$ for values of x from -3 to 3



(2)

24. 4MA1/2HR_Winter_2022_Q23

The functions f and g are such that

$$f(x) = x + 25 \qquad g(x) = x^2 - 12x$$

The function h is such that $h(x) = fg(x)$

The domain of h is $\{x : x \leqslant 6\}$

Express the inverse function h^{-1} in the form $h^{-1}(x) = \ldots$

$h^{-1}(x) =$ ....

4 marks

25. 4MA1/2HR_Winter_2022_Q18

The line with equation $2y = x + 1$ intersects the curve with equation $3y^2 + 7y + 16 = x^2 - x$ at the points A and B

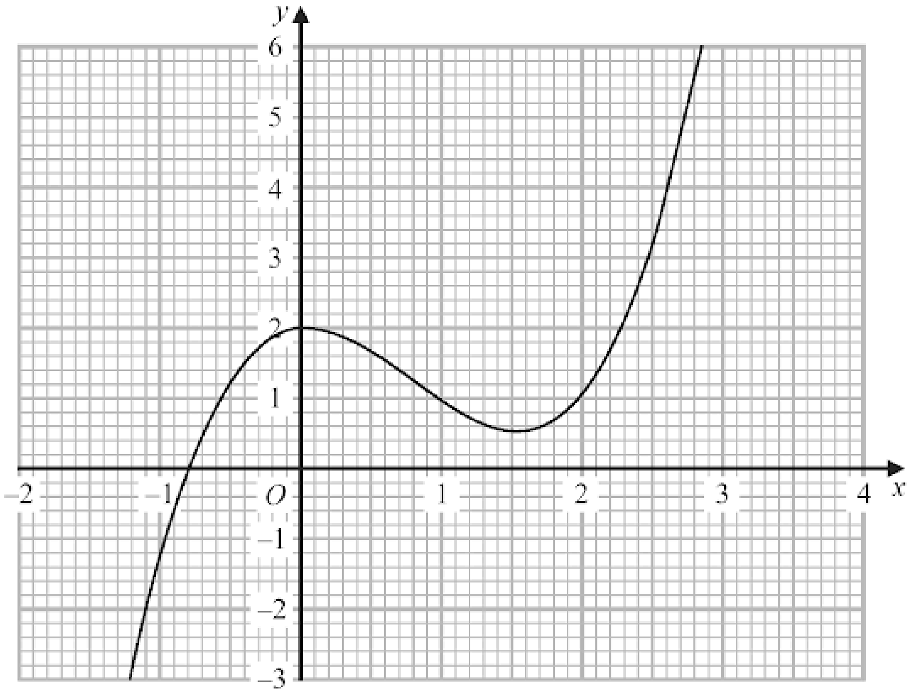
Find the coordinates of A and the coordinates of B
Show clear algebraic working.

(.................... , ...) and (.................... , ...)

5 marks

26. 4MA1/2HR_Winter_2022_Q17

Part of the curve with equation $y = \mathrm{f}(x)$ is shown on the grid.



Find an estimate for the gradient of the curve at the point where $x = 2$
Show your working clearly.

....

3 marks

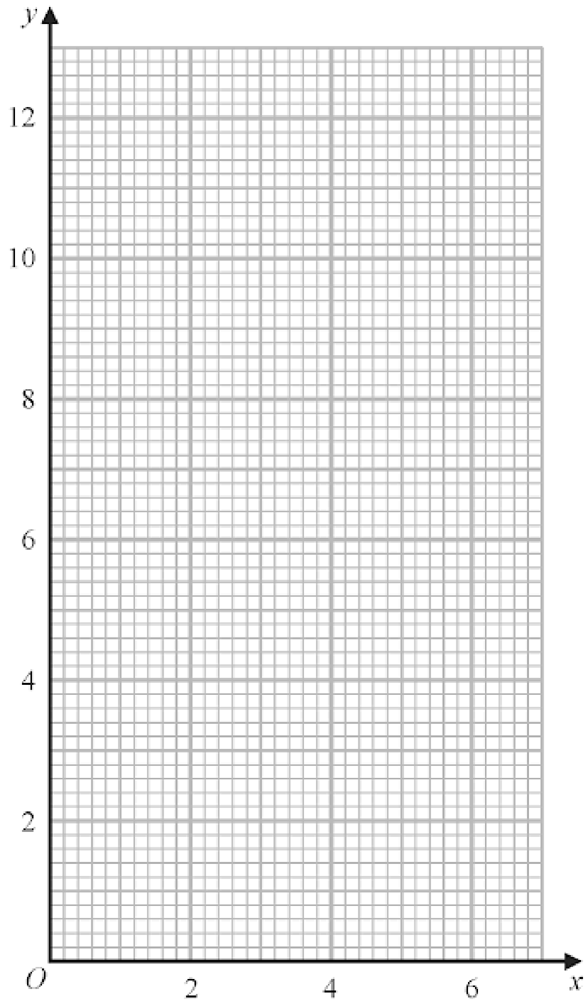
 4MA1/2HR_Winter_2022_Q11

(a) Complete the table of values for $y = \dfrac{6}{x}$

x	0.5	1	2	3	4	5	6
y		6		2			1

(2)

(b) On the grid, draw the graph of $y = \dfrac{6}{x}$ for $0.5 \leqslant x \leqslant 6$



(2)

28. 4MA1/2HR_Winter_2022_Q7

(a) Write down an equation of the straight line with gradient -3 and which passes through the point with coordinates $(0, 5)$

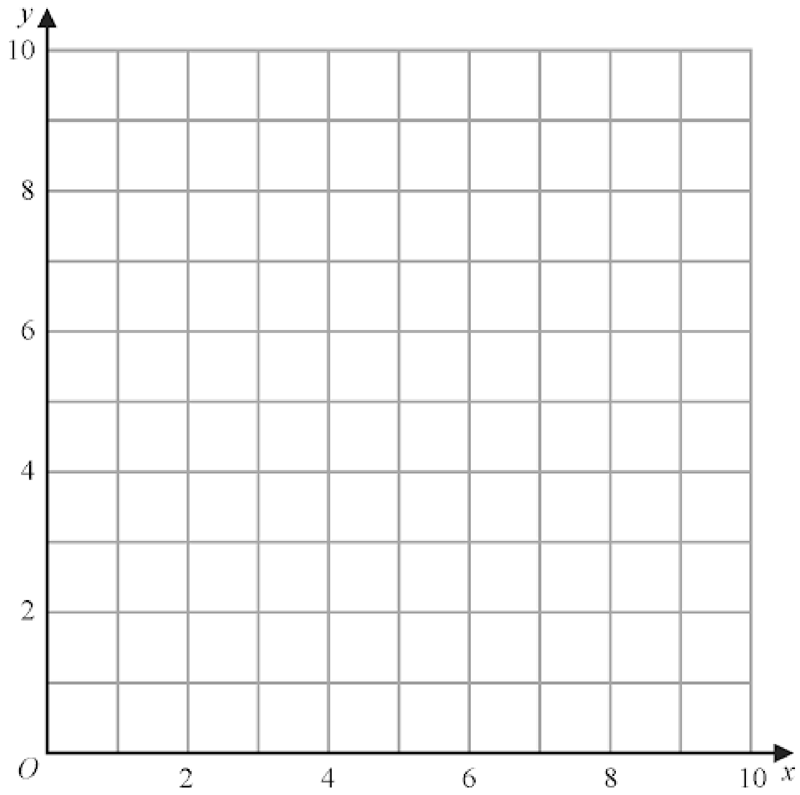
....

(2)

(b) Show, by shading on the grid, the region defined by **all three** of the inequalities

$x \leqslant 6$ $\qquad$ $y \geqslant 2$ $\qquad$ $y \leqslant x + 1$

Label the region **R**



(3)

29. 4MA1/2H_Winter_2022_Q26

An arithmetic series has first term a and common difference d, where d is a prime number.

The sum of the first n terms of the series is S_n and

$S_m = 39$

$S_{2m} = 320$

Find the value of d and the value of m
Show clear algebraic working.

$d =$ ....

$m =$ ....

5 marks

30. 4MA1/2H_Winter_2022_Q25

The function g is defined as

$$g : x \mapsto 5 + 6x - x^2 \quad \text{with domain } \{x : x \geqslant 3\}$$

(a) Express the inverse function g^{-1} in the form $g^{-1} : x \mapsto \ldots$

$g^{-1} : x \mapsto$ ....

(4)

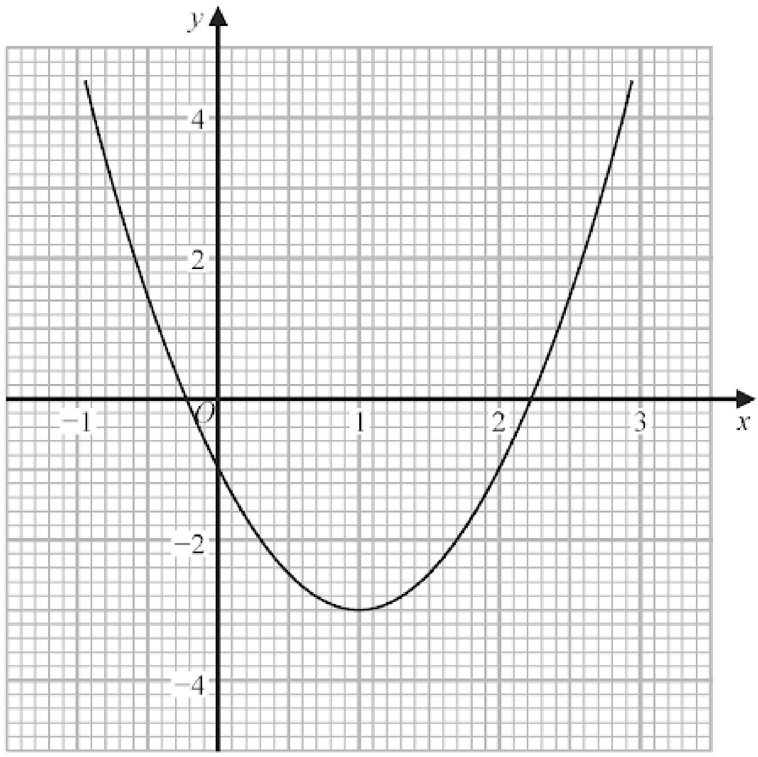
(b) State the domain of g^{-1}

....

(1)

31. 4MA1/2H_Winter_2022_Q21

Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



(a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

....

(2)

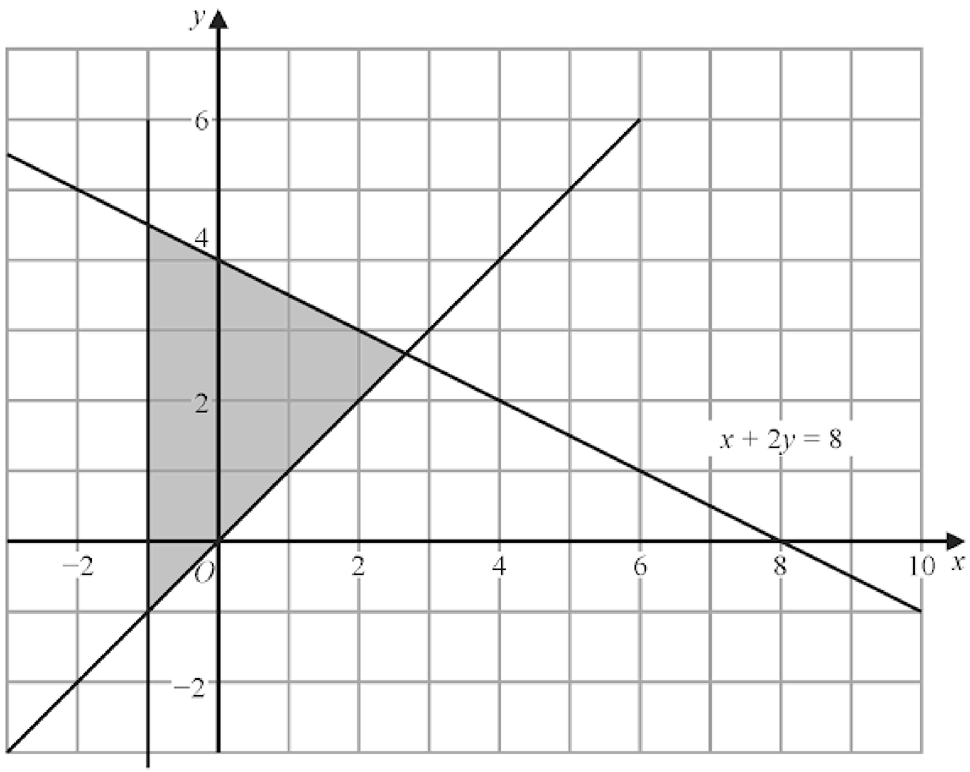
(b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

....

(3)

32. 4MA1/2H_Winter_2022_Q13

The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down three inequalities that define the shaded region.

....

....

....

3 marks

The point A with coordinates $(-3, 2)$ lies on the straight line with equation $y = \mathrm{f}(x)$

(a) Find the coordinates of the image of the point A on the straight line with equation

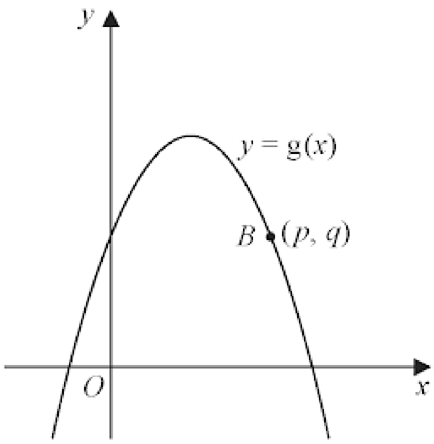
(i) $y = \mathrm{f}(x) - 3$

(.................... , ...)
(1)

(ii) $y = \mathrm{f}\left(\frac{x}{2}\right)$

(.................... , ...)
(1)

Here is a sketch of part of the curve with equation $y = \mathrm{g}(x)$



The point B with coordinates (p, q) lies on the curve.

(b) Find the coordinates of the image of the point B on the curve with equation

$$y = -\mathrm{g}(x - c)$$

where c is a constant.

(.................... , ...)
(2)

34. 4MA1/2HR_Summer_2022_Q19

$\mathrm{f}(x) = x^2 - 4$

$\mathrm{g}(x) = 2x + 1$

Solve $\mathrm{fg}(x) > 0$

Show clear algebraic working.

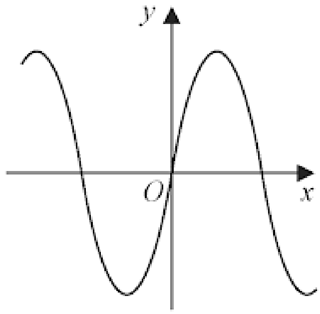
....

4 marks

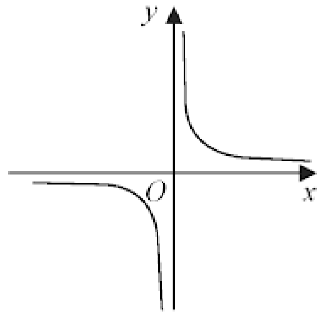
35. 4MA1/2HR_Summer_2022_Q16

Here are nine graphs.

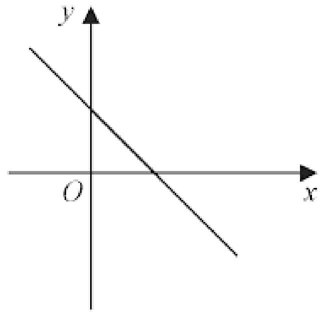
Graph A



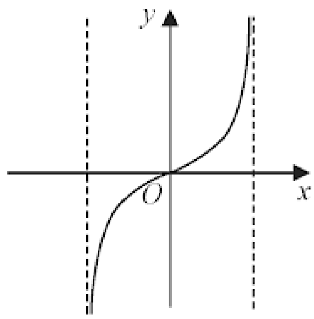
Graph B



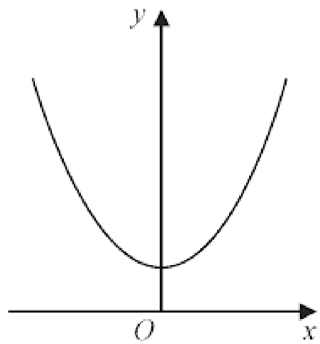
Graph C



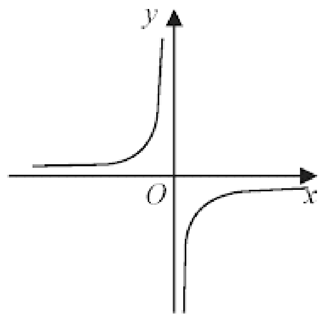
Graph D



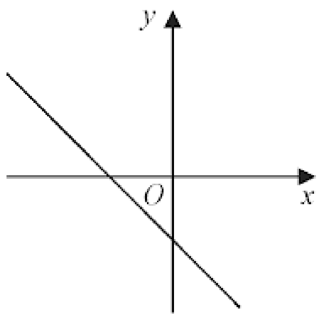
Graph E



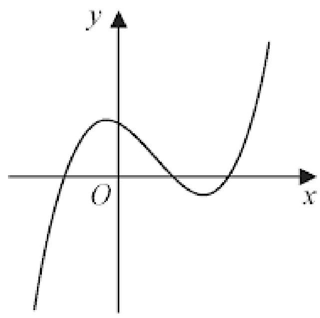
Graph F



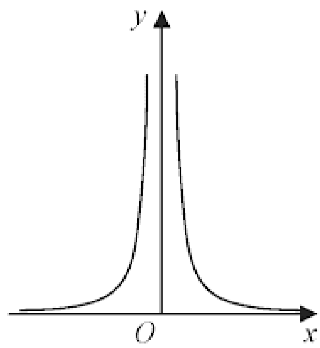
Graph G



Graph H



Graph I



36. 4MA1/2H_Summer_2022_Q22

$ABCD$ is a kite, with diagonals AC and BD, drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

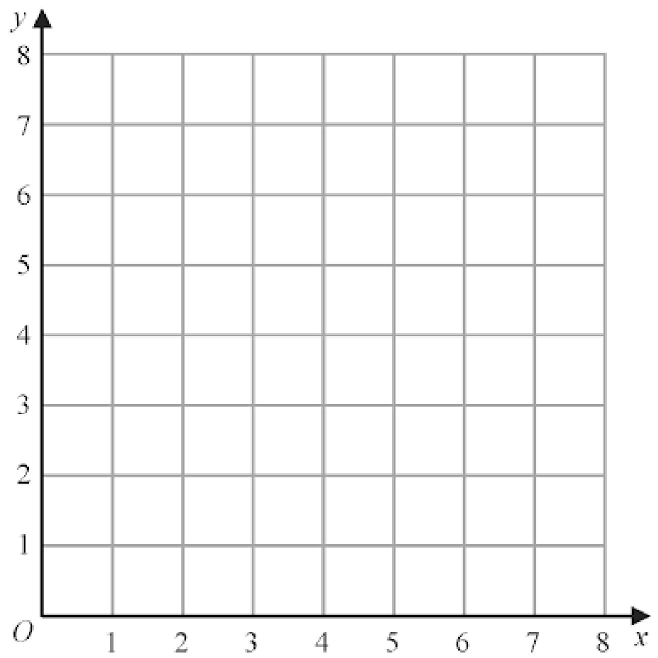
Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,

find the coordinates of the points B and D.

37. 4MA1/2H_Summer_2022_Q2

(a) On the grid, draw and label with its equation the straight line with equation

(i) $y = 1$ (ii) $x = 2$ (iii) $x + y = 7$



(3)

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$y \geqslant 1$ $x \geqslant 2$ $x + y \leqslant 7$

Label the region **R**.

(1)

38. 4MA1/2HR_Winter_2021_Q17

The functions f and g are defined as

$$f(x) = x^2 + 6$$
$$g(x) = x - 10$$

(a) Find fg(3)

....

(2)

(b) Solve the equation fg(x) = f(x)
Show clear algebraic working.

....

(3)

The function h is defined as

$$h(x) = \frac{2x - 4}{x}$$

(c) State the value of x that cannot be included in the domain of h

....

(1)

(d) Express the inverse function h^{-1} in the form $h^{-1}(x) = \ldots$

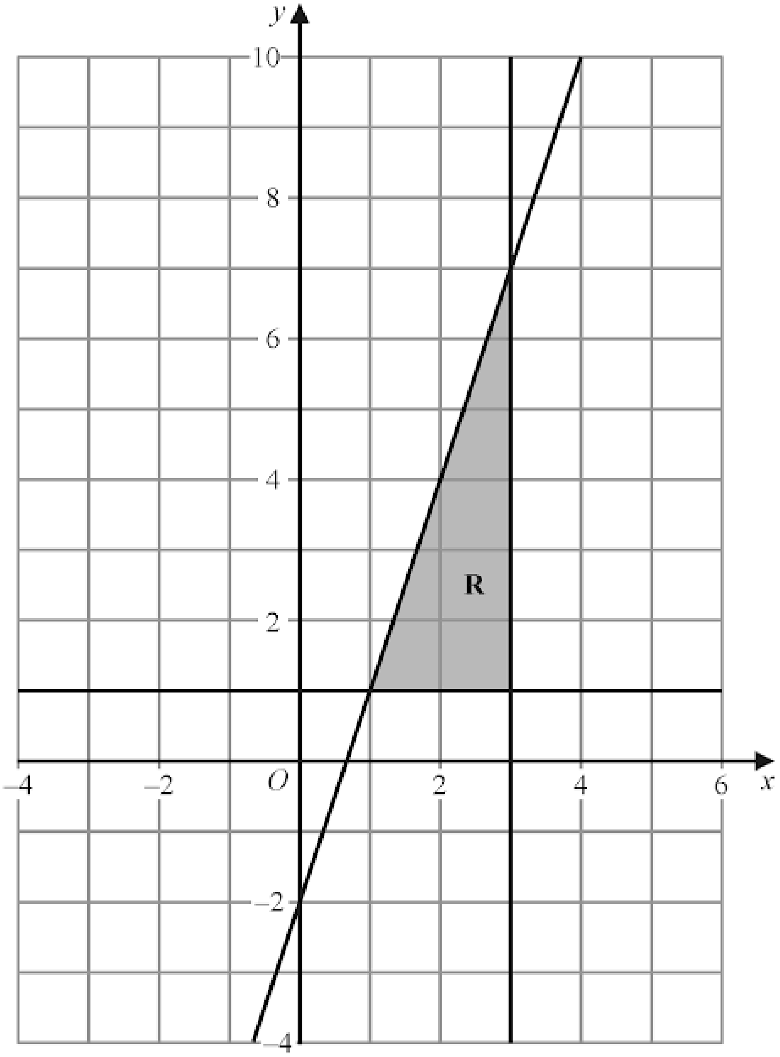
$h^{-1}(x) =$ ....

(3)

39. 4MA1/2HR_Winter_2021_Q7

The shaded region **R**, shown in the diagram below, is bounded by the straight line with equation $y = 3x - 2$ and by two other straight lines.

Write down the three inequalities that define region **R**.



....

....

....

(Total for Question is 3 marks)

40. 4MA1/2H_Winter_2021_Q18

A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$
B has coordinates $(1, 10)$
C has coordinates $(19, a)$
D has coordinates (b, c)

(a) Work out the value of a, the value of b and the value of c.

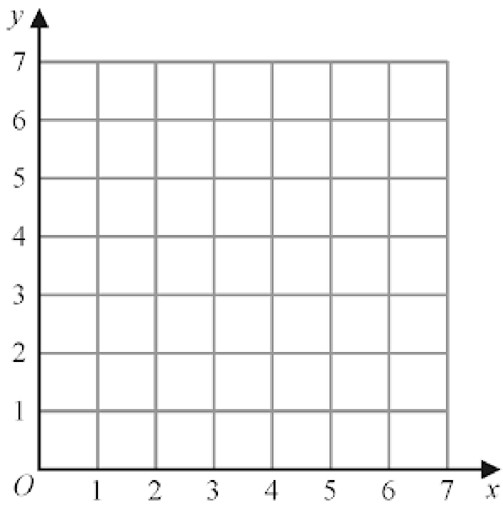
$a =$ ....

$b =$ ....

$c =$ ....

(4)

41. 4MA1/2H_Winter_2021_Q4



(a) On the grid, draw and **label** the straight line with equation

(i) $x = 1.5$

(ii) $y = x$

(iii) $x + y = 6$

(3)

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$x \geqslant 1.5 \qquad y \geqslant x \qquad x + y \leqslant 6$

Label the region **R**.

(1)

42. 4MA1/2H_Summer_2021_Q22

The curve **S** has equation $y = f(x)$ where $f(x) = x^2$
The curve **T** has equation $y = g(x)$ where $g(x) = 2x^2 - 12x + 13$

By writing $g(x)$ in the form $a(x - b)^2 - c$, where a, b and c are constants, describe fully a series of transformations that map the curve **S** onto the curve **T**.

....

....

....

4 marks

43. 4MA1/2H_Summer_2021_Q16

A is inversely proportional to the square of r

$A = 5$ when $r = 0.3$

(a) Find a formula for A in terms of r

....

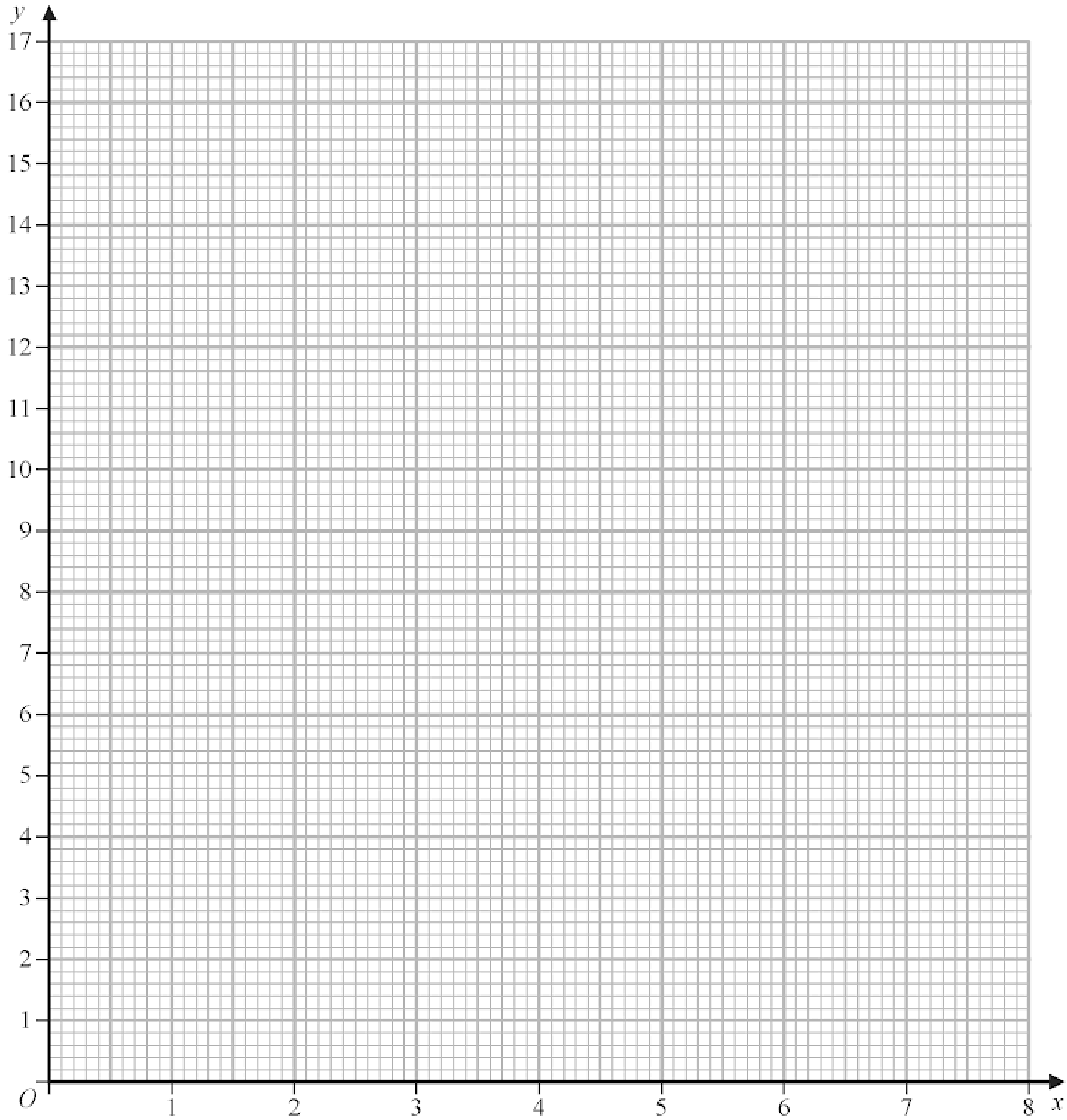
(3)

(b) Find the value of A when $r = 7.5A$

$A =$ ....

(3)

(b) On the grid, draw the graph of $y = \frac{1}{x}(x^2 + 4)$ for $0.25 \leqslant x \leqslant 8$



(2)

45. 4MA1/2HR_Winter_2020_Q24

$\mathbf{L}_1$ and $\mathbf{L}_2$ are two straight lines.
The origin of the coordinate axes is O.

$\mathbf{L}_1$ has equation $5x + 10y = 8$
$\mathbf{L}_2$ is perpendicular to $\mathbf{L}_1$ and passes through the point with coordinates (8, 6)

$\mathbf{L}_2$ crosses the x-axis at the point A.
$\mathbf{L}_2$ intersects the straight line with equation $x = -3$ at the point B.

Find the area of triangle AOB.
Show your working clearly.

....

(Total for Question is 5 marks)

46. 4MA1/2HR_Winter_2020_Q23

Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a, b and c are integers.

....

(Total for Question is 3 marks)

47. 4MA1/2HR_Winter_2020_Q22

The first term of an arithmetic series S is -6
The sum of the first 30 terms of S is 2865

Find the 9th term of S.

....

(Total for Question is 4 marks)

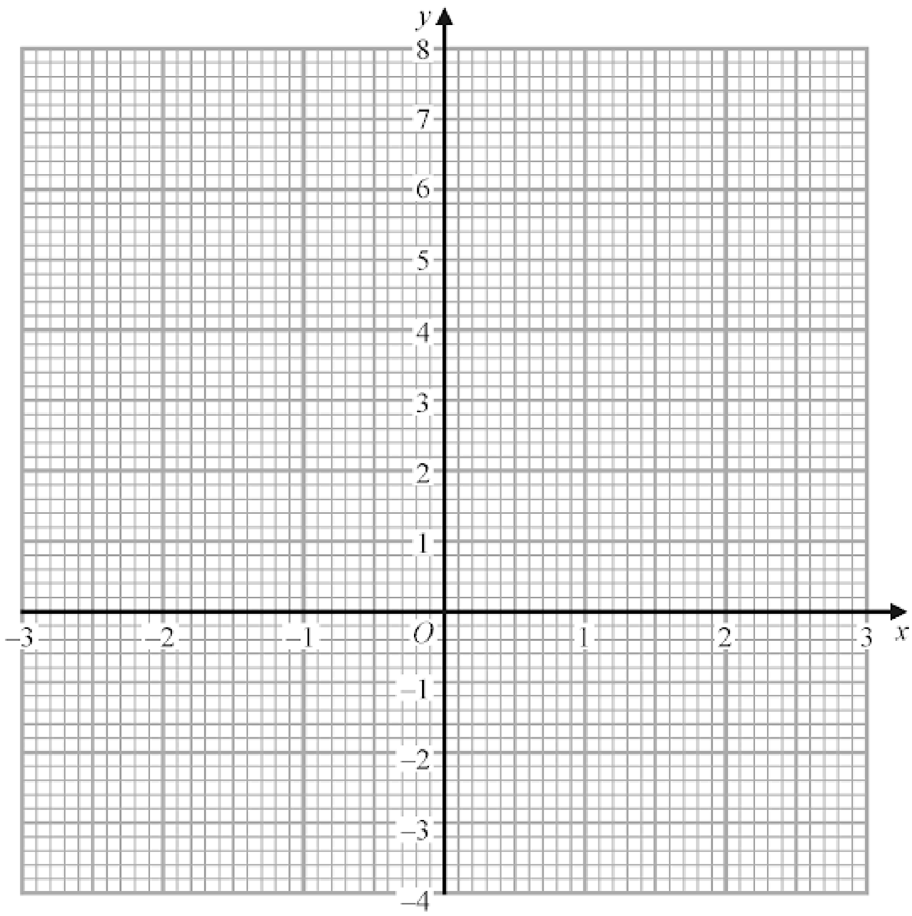
48. 4MA1/2HR_Winter_2020_Q15

(a) Complete the table of values for $y = x^2 - \frac{x}{2} - 3$

x	–3	–2	–1	0	1	2	3
y	7.5				–2.5		4.5

(2)

(b) On the grid, draw the graph of $y = x^2 - \frac{x}{2} - 3$ for values of x from –3 to 3



(2)

49. 4MA1/2HR_Winter_2020_Q11

(a) Write down the integer values of x that satisfy the inequality $-2 < x \leqslant 4$

....
(2)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.

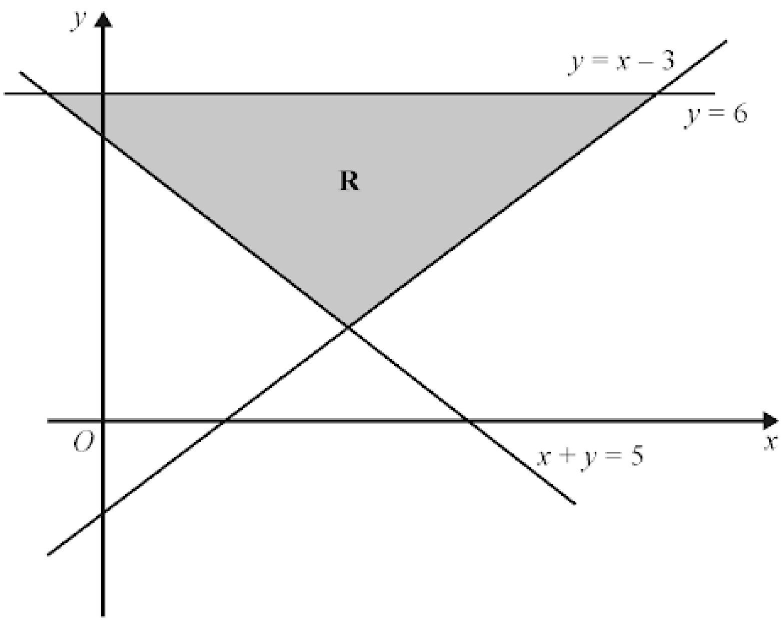


Diagram **NOT** accurately drawn

(b) Write down the three inequalities that define the region **R**.

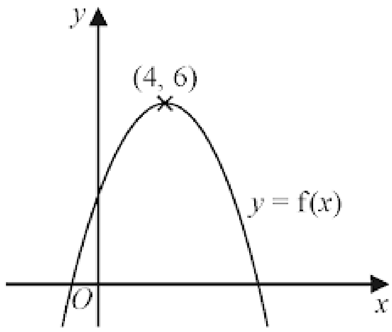
....

....

....
(2)

50. 4MA1/2H_Winter_2020_Q21

The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.
The coordinates of this maximum point are (4, 6)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 4)$

(.................... , ...)

(ii) $y = f(2x)$

(.................... , ...)

(2)

The equation of a curve **C** is $y = x^2 + 3x + 4$

The curve **C** is transformed to curve **S** under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$

(b) Find an equation of curve **S**.

You do not need to simplify the equation.

....

(2)